

AZIZ SHAMEEM

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EDUCATION

Indian Institute of Technology Bombay

B.Tech in Electrical Engineering and M.Tech in Artificial Intelligence (**Cumulative GPA: 9.71/10.0**)

Mumbai, India

Nov'20 – Present

PUBLICATIONS

- ♦ **A. Shameem**, A. Sethi, “**Better Sampling as a Path to Stronger VAEs**”, ICLR 2025 DeLTa Workshop, 2025
- ♦ M. Shukla, **A. Shameem**, M. Salzman, A. Alahi, “**Towards Self-Supervised Covariance Estimation in Deep Heteroscedastic Regression**”, ICLR 2025
- ♦ V. Prasad, **A. Shameem**, C. White, S. Nayak, P. Jain, P. Garg, G. Ramakrishnan, “**Speeding up NAS with Adaptive Subset Selection**”, AutoML 2024

RESEARCH EXPERIENCE

Variational Auto-Encoders and Diffusions Models

May'24 – Present

Summer Internship, Guide: [Prof. Alexandre alahi](#)

EPFL

- Performed extensive analysis of the **similarities and differences** between two types of Generative Frameworks - **Variational AutoEncoders** and **Diffusion Models**
- Designed experiments to elucidate the **limitations** of VAEs, and suggested **improvements/methods** to overcome them
- Demonstrated methods for better generations by VAEs on **synthetic** as well as **benchmark datasets**
- Incorporated the methods for improving **conditional image generation using VAEs**
- Introduced **Probabilistic Optimizations** to better the computational complexity of the algorithms

OPAS : Ordered Permutation-based Alignment for Sequence matching

Jun'23 – Oct'24

Guide: [Prof. Abir De](#)

CSE, CMInDS, IIT Bombay

- Developed a **theoretical framework** based on **lagrangians** for **fast, approximate, constrained sequence matching**
- Tested the developed framework **computationally**, and obtained results on datasets of several **modalities**
- Compared the developed method with several baselines, and demonstrated over **58x speedup** over them
- Compared several **implementation variants, architectures and models** of the proposed system as ablations

Neural Architecture Search and Data Subset Selection

May'23 – Oct'23

Guide: [Prof. Ganesh Ramakrishnan](#)

CSE, IIT Bombay

- Enhanced existing **pruning**-based Neural Architecture Search, experimenting with proven pruning techniques
- Incorporated **adaptive subset selection** to enhance the efficient use of data and obtained competitive results in the same

Adversarial Domain Adaptation

Jan'23 – May'23

Guide: [Prof. Amit Sethi](#)

EE, IIT Bombay

- Developed the complete codebase for fitting a novel model - **WaveMix** for use in a task involving **Domain Adaptation**
- Enhanced the performance of the network to increase its **accuracy** and **generalisability** to unseen datapoints

PROFESSIONAL EXPERIENCE

Modem Systems Engineering Intern

May'23 – Jul'23

Qualcomm Technologies Inc.

Bengaluru, India

- Undertook extensive research on enhancing **information-deficit channel estimation** methods for **5G**
- Implemented existing as well as proposed technology in a **python simulator** and verified theoretical results
- Implemented specific **channel models** in the python simulator, to be used for future research and development

SCHOLASTIC ACHIEVEMENTS

- Current holder of **Department Rank 1** in **Centre for Machine Intelligence and Data Science** (Present)
- Ranked **1st** in Electrical Engineering Dept. (Dual Degree) out of a batch of 100 students for two years (2022-24)
- Recipient of the **Summer@EPFL Travel Scholarship** (awarded to 1.6% of the applicants) (2024)
- Awarded **Institute Academic Prize** for exceptional performance in academics for **two consecutive years** (2022, 2023)
- Secured an **All India Rank of 62** in JEE Mains (Engineering) among 1.3 million candidates (2020)
- Secured an **All India Rank of 665** in JEE Advanced among 0.2 million candidates (2020)
- Conferred with **AP grade (Advanced Performer)** in **Advanced ML, Linear Algebra, Electronic Devices**
- Ranked in the **national top 1%** in NSEC and selected to appear for Indian National Chemistry Olympiad (2019)

KEY TECHNICAL PROJECTS

Auto-Grad Engine and GPT development

May'23 – Jul'23

Self Project

IIT Bombay

- Built an **Auto-Grad Engine** having **pytorch-like API** from scratch and demonstrated its working on a test bed
- Performed extensive study on **innate problems** in neural networks pertaining to **gradients and dead neurons**
- Implemented initialization and normalization based methods to bypass problems and demonstrated better results
- Developed a working **GPT-like architecture** from scratch and generated recognizable **Shakespearean text**

Toxicity Detection in Large Language Models

Apr'24 – May'24

Guide: [Prof. Pushpak Bhattacharya](#)(CS772 : DL for NLP)

IIT Bombay

- Trained a **text toxicity classifier** using **Recurrent Neural Nets, LSTMs** and **Transformers**
- **Engineered additional features** from the analysis of textual data in an effort to improve classification of toxicity in texts
- Utilised the trained classifier network as a **filter** to mitigate toxicity in the output of **Large Language Models**

Out of Distribution Detection

Apr'23 – May'23

Guide: [Prof. Sunita Sarawagi](#)(CS726 : Advanced ML)

IIT Bombay

- Performed a literature survey on **energy-based Out-Of-Distribution Detection** on Computer Vision and Image Datasets.
- Replaced the proposed loss function with a novel **Ranking Loss**, and displayed its superior performance.
- Proposed changes to the documented architecture in an effort to apply the study to **text datasets**

Correlated Multi-Armed Bandits

Mar'23 – Apr'23

Guide: [Prof. Jayakrishnan Nair](#), [Prof. D. Manjunath](#) (EE6106: Online Learning and Optimization)

IIT Bombay

- Performed an extensive literature review on the formulation of the **correlated multi-armed Bandit** problem
- Applied an existing framework for the **exploitation of correlation** between bandits, on different algorithms and demonstrated **substantial improvements** in the same
- Designed effective visualizations for **comparing** the implemented algorithms and **identification** of the **best arm**

Music Genre Classification using ML and DL

Apr'22 – May'22

Guide: [Prof. Biplob Bannerjee](#)(DS303: Introduction to Machine Learning)

IIT Bombay

- Implemented **Decision Trees, Random Forest, Naive Bayes, SVM** and **KNN** to classify music based on its genre
- Tried several architectures and hyper-parameters of **Sequential Neural Networks** to classify music based on genre

Machine Learning for COVID-19 Data Analysis

Oct'21 – Nov'21

Guide: [Prof. Amit Sethi](#)(DS203: Programming for Data Science)

IIT Bombay

- Performed extensive **Exploratory Data Analysis** on **COVID-19** in India and several other countries across two years
- Used **Linear, Polynomial, Lasso, Ridge Regression models** to predict future COVID-19 related casualties
- Implemented **Logistic Regression, SVM, NN, RF** and **GBC** to predict need of ICU admission based on medical factors

Gas Leakage Detection and VSLAM using Nanosaur

Jan'23 – Apr'23

Guide: [Prof. Siddharth Tallur](#) (EE344 : Electronic Design Lab)

IIT Bombay

- Ideated and designed a mobile robot capable of **toxic gas detection** and **3D reconstruction** of its environment
- Carried out the design and testing of the **Printed Circuit Board** for interfacing the gas sensors with Jetson Nano
- Created **ROS Nodes** for efficient **sensor data accumulation and display**, and got **recognized** for the same

Expense Tracker

May'21 – Jul'21

Python is Cool, Kids (PyCK) : Summer Course

IIT Bombay

- Developed an **application** capable of storing and plotting users' expenses in **Python**
- Created an interactive **Front-End** for the project to enable **account creation** and **data entry** into the system using tkinter
- Used Python Libraries **Pandas** and **Matplotlib** to create a **Back-End** to store and revert data as and when required

OTHER PROJECTS

- Carried out Extensive Data Analysis of **IPL**, and developed systems to predict **scores and outcomes** of matches (2021)
- Created basic games like **zig-zag** and **candy-cush** using **Unity and C#**, as part of an online course on GameDev (2021)
- Designed and simulated the Schematic and layout of common Amplifiers and Ring oscillator on Cadence (2022)
- Designed and implemented the **datapath** and **control logic** for a **16-bit multi-cycle processor** in HDL (2022)
- Designed and Implemented a **PID controller** for a **Line Following Robot** as part of Control Systems Laboratory (2022)

TECHNICAL SKILLS

Data Analysis/Visualization tools : Pandas, Numpy, Matplotlib, seaborn, plotly, MATLAB, MySQL

Libraries: Pytorch, PyTorch-Geometric, Tensorflow, Numpy, Pandas, SciPy, Seaborn, Scikit-Learn, OpenCV, Matplotlib

Software: Git, ~~La~~TeX, MATLAB, OpenCV, GNU Radio, Quartus, NGSpice, Kicad, Arduino, Pycharm, Anaconda, Unity

POSITIONS OF RESPONSIBILITY

Machine Learning Engineer

Data Analysis and Visualisation Team, IITB

Jun'22 – Apr'23

- Led a team to **successful completion and publication** (On the official public handle) of analysis on **Indian Establishments and Settlements**
- Performed extensive analysis on **Grading Statistics** and **Semester Exchange statistics** of the institute and oversaw its **publication on LinkedIn**
- Implemented the **YOLO version-1** framework from scratch as part of a study project and gave a presentation on the same
- Represented team at a kaggle competition which involved **data pre-processing, model design and training** on NLP tasks
- Delivered a comprehensive presentation on **Loss Functions for Image Restoration** and got recognized for the same

Mentor : Introduction to Machine Intelligence

Seasons of Code, IITB

Apr'22 – Jul'22

- Prepared and distributed **comprehensive resources** to help beginners understand and appreciate concepts in ML
- Assisted beginners in building trivial **AI systems** using **Reinforcement Learning** in an effort to generate interest in the field

Junior Control Systems Engineer

Hyperloop, IITB

Oct'21 – Apr'22

- Learned and implemented **sensor data collection** algorithms in MATLAB, based on **Kalman Filter**
- Studied and applied **sensor fusion** to efficiently collect and utilize in-flowing data from designated sensors
- Learned and applied concepts on **PCB design** and **device interfacing** to build compact controllers

Web Developer/Coordinator

Techfest - Asia's largest science and technology festival

Oct'21 – Feb'22

- Contributed to the creation of **TechFest Website** by developing **hover animations, Bootstrap cards and Carousels**
- Improvised and optimised existing code-base by appropriate fixes and restructuring

KEY COURSEWORK

Machine Learning	Advanced ML Deep Learning for NLP Learning with Graphs Mathematical Optimization Techniques Optimization in ML Online Learning and Optimization Introduction to Machine Learning
Computer Science	Programming for Data Science Image Processing Design and Analysis of Algorithms Data Structures and Algorithms Digital Systems Microprocessors Computer Programming and Utilization
Math and Probability	Markov Chains and Queueing Processes Applied Linear Algebra Advanced Probability and Random Processes Matrix Computations Estimation and Identification
Online Certifications	Reinforcement Learning Specialization Python for Computer Vision and Deep Learning Machine Learning Unity C# Scripting

EXTRA CURRICULAR ACTIVITIES AND OTHER ACHIEVEMENTS

TAship	Complex Analysis Linear Algebra	Introduction to Machine Learning Programming for Data Science
Chess	Secured First Place in 9-Up , a tournament organized by Hostel-9 sports council Represented Hostel 9 in the General Championships at IITB for two consecutive years Received a trophy for Honourable Mention at the 1st Tarabai Shinde Chess Tournament	
Coding	Stood third in an Algorithmic Trading hackathon organised by Web and Coding club, IITB	
Tennis	Represented Hostel 9 in the General Championships at IITB for two consecutive years	
Miscellaneous	Appointed as the Class Representation of the CMInDS IDDDP batch of 2020 Regular participant of the 5km inter-hostel marathon runs conducted twice a year	